

Multiple Choice

1. **Answer:** E

Options: A) By using a controlled group of dogs not exposed to the sounds; B) By using sensory deprivation to heighten hearing sensitivity; C) By observing the dog's natural behavior in different sound environments; D) By using negative reinforcement only; E) By using operant conditioning

Note: Operant conditioning allows researchers to assess a dog's sensitivity to sound frequencies by reinforcing specific responses to different auditory stimuli, thereby measuring the dog's ability to detect and discriminate between varied frequencies.

2. **Answer:** E

Options: A) Enzyme-linked immunosorbent assay (ELISA); B) Polymerase chain reaction (PCR); C) X-ray crystallography; D) Mass spectrometry; E) Autoradiography

Note: Autoradiography allows researchers to trace the incorporation of labeled amino acids into newly synthesized proteins, demonstrating the role of amino acids in protein synthesis while also indicating the breakdown of existing proteins through the visualization of radioactivity patterns.

3. **Answer:** A

Options: A) The thick muscular wall of the stomach; B) Antibacterial properties of stomach acid; C) Gastric juice; D) High pH levels in the stomach; E) Presence of beneficial bacteria

Note: The thick muscular wall of the stomach provides a protective barrier that prevents the gastric acids and enzymes from digesting the stomach tissue itself, allowing it to contain and process food safely.

4. **Answer:** D

Options: A) Conjugation is prevented by the presence of a dominant allele; B) Conjugation results in lesser genetic diversity; C) Conjugation can occur among identical organisms; D) All individuals are identical genetically, and conjugation could not occur.; E) Conjugation does not require different mating types

Note: Conjugation in paramecia requires genetic diversity between individuals, and if all individuals descended from a single individual through repeated fission, they would be genetically identical, making conjugation impossible.

5. **Answer:** B

Options: A) the enzyme is inactive at all pH values; B) no ionizing groups on the enzyme or substrate participate in the reaction; C) only ionizing groups on the substrate participate in the reaction; D) ionizing groups on both the enzyme and substrate participate in the reaction; E) only non-ionizing groups on the substrate participate in the reaction

Note: The correct answer is based on the fact that if an enzyme's activity remains constant across various pH levels, it indicates that the reaction does not rely on ionizable groups that would be affected by changes in pH, suggesting that either the enzyme or substrate lacks such groups that influence catalytic activity.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Aposematic coloration refers to warning coloration that signals to predators that an organism is toxic or unpalatable, rather than using camouflage to avoid detection.

7. **Answer:** True

Options: A) True; B) False

Note: The ability of the brain to detect differences in stimulus intensity is influenced by the number of synapses crossed by sensory signals because each synapse can modulate the strength and integration of the signals, impacting how differences in intensity are processed and perceived.

8. **Answer:** True

Options: A) True; B) False

Note: The synthesis of an RNA/DNA hybrid from a single-stranded RNA template is true because it necessitates the action of reverse transcriptase, which requires a primer to initiate the synthesis of complementary DNA from the RNA template.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because molting in crustaceans is primarily regulated by hormones such as ecdysteroids and molt-inhibiting hormones, which coordinate the timing and process of shedding the exoskeleton, independent of external factors like water temperature.

10. **Answer:** False

Options: A) True; B) False

Note: Hemoglobin's affinity for O₂ actually decreases as blood pH decreases, a phenomenon known as the Bohr effect, which facilitates oxygen release to tissues during conditions of higher carbon dioxide and lower pH.

Long Form Response

11. **Answer:** Water availability plays a crucial role in the transition from forest ecosystems to grasslands across the U.S. as one moves from east to west. The eastern regions, characterized by higher precipitation levels, support denser forests due to adequate moisture for tree growth and biodiversity. However, as one travels westward, decreasing rainfall leads to drier conditions, which limit the growth of tall trees and promote

the dominance of grasses and herbaceous plants. This shift in vegetation is a direct response to the availability of water, where reduced moisture availability favors grassland ecosystems that are more resilient to drought and can thrive in less fertile soils. Thus, water availability is a key factor that shapes the distinct ecological zones across the continent.

Note: correct because it accurately describes how the gradient of water availability influences vegetation types, resulting in dense forests in the wetter eastern regions and the prevalence of grasslands in the drier western areas.

12. **Answer:** Kinesis and taxis are two distinct forms of animal behavior that describe how organisms respond to stimuli in their environment. Kinesis refers to a non-directional movement where an animal alters its activity level or speed in response to a stimulus, but does not orient its body toward it; for example, woodlice increase their movement in dry conditions to find moisture. In contrast, taxis involves a directional movement where an organism moves toward or away from a specific stimulus, such as moths flying toward a light source at night. These behaviors illustrate different strategies animals use to navigate their environments, with kinesis providing a more random response and taxis offering a targeted approach. Understanding these concepts is crucial in studying how animals adapt and survive in various ecological contexts.

Note: correct because it accurately distinguishes between kinesis as a non-directional response characterized by changes in activity level and taxis as a directional movement toward or away from a stimulus, while providing relevant examples that illustrate each concept in natural settings.